

BS QUIZ

MBD

MBD

model based definition: 3d cad model becomes primary data source for manufacturing data, tolerances, inspections, materials, etc. → eliminates 2d dwgs for documentation

all other documents derive from MBD models

MBE

model based enterprise: an environment that uses and shares MBD based models across different teams and departments

PMI:

product & manufacturing information: data contained in MBDs like materials, datums, tolerances, surface finish, tolerance stack up, CMM analysis, etc.

PMI >> MBD >> MBE (SCALE)

Benefits:

increased efficiency: lesser need for 2d dwgs, time saved to create and manage info

greater accuracy: only one common source for data

improved collaboration: product data is easily shared between teams

reduced costs: less cost needed to create, manage and distribute dwgs

better quality: accurate and consistent information shared

greater flexibility: information can be easily changed and distributed

improved sustainability: due to increased efficiency

Detailing

?????

Cams

Cam terms

Base Circle

smallest circle of cam tangent to the cam's surface

Lift/Stroke

the distance travelled by the follower

trace point

point on the follower that makes a pitch curve around the cam.

for knife edge follower, the trace point is the point of contact with the cam

for roller followers, the trace point is the center of the roller

pitch curve

curve generated by the trace point

Pressure angle

angle between line of action and direction, varies

pitch point

point where pressure angle is a maximum

pitch circle

circle formed by pitch point and center of cam

types of cams

disk/plate

cylindrical

translating

wedge cam

spiral cam

heart shape cam

face cam

types of followers

by shape:

knife edge, roller, flat face, spherical

by location:

radial follower, offset follower

by motion:

reciprocating (linear), oscillating (about an axis)

pros and cons

pros: simple to use, easily able to program complex motion

cons: high wear and tear, not suitable for high loads or speeds

design considerations

loading conditions

cam & follower should be able to handle radial & axial loads

motion considerations

higher speeds $\Rightarrow$ lower friction, higher precision needed; acceleration and deceleration also influences forces and design

follower type

in general,

knife edge provides higher precision, but have higher wear under faster speeds

roller followers provide lower friction, higher load and shock absorption, but lesser accuracy

flat face followers provide balance of load capacity and precision

lubrication & maintenance

lubrication provisions like grease fittings and oil reservoirs should be accommodated for reducing friction and heat, based on operating conditions and maintenance intervals

material selection

based on load capacity, operating conditions, and environmental factors

hardened steel, stainless steel, or other alloys commonly used

application based

dependent on application, like sound requirements, space limitations, etc.

Frames & Weldments

Types of frames

frames: structures consisting of columns and beams (vertical and horizontal)

can be either rigid (fixed joints) or flexible (pinned/movable joints)

fixed joints may be done through welding, pinned joints done by riveting

Aluminium Extrusion frames

benefits:

- cheap

- high formability & easy to form into complex geometries

- high strength to weight ratio

- corrosion resistant

- wide range of surface finishes (anodizing, sand blasting, etc.)

frames can be connected with brackets or insertion nuts

Weldments

Weld Joint Types

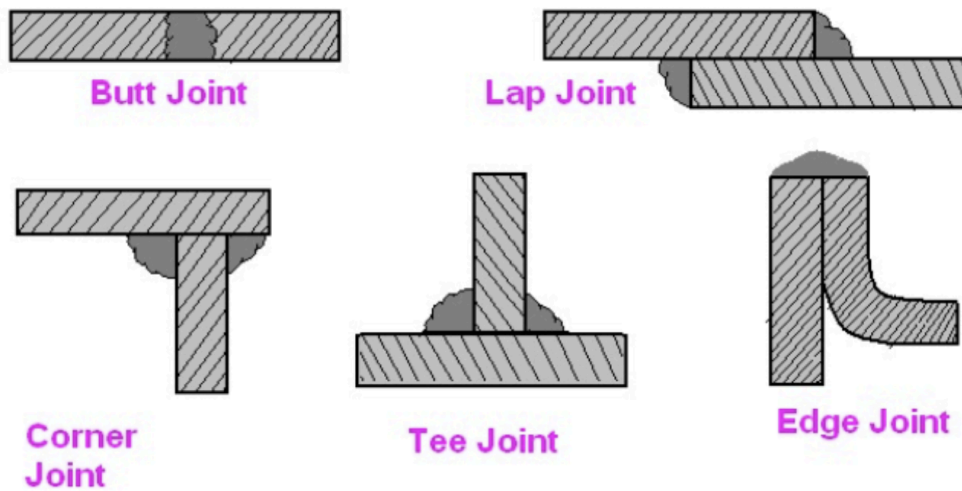
- Butt Joint

- Lap Joint

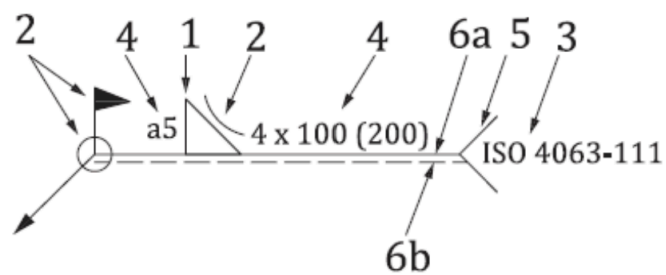
- Corner Joint

- Tee Joint

- Edge Joint

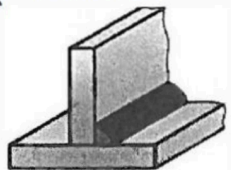
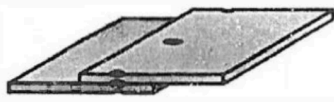


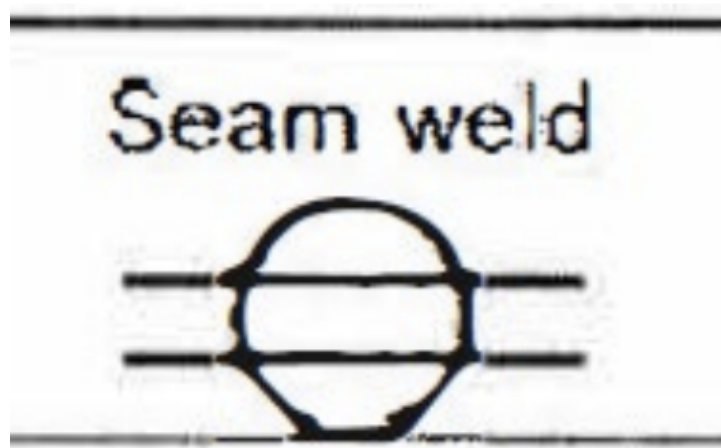
Weld Symbol

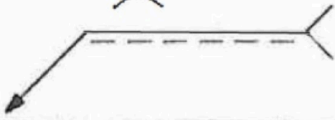
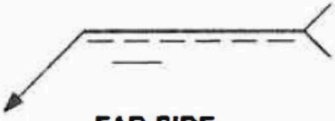


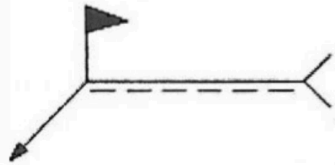
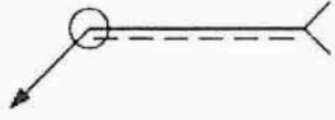
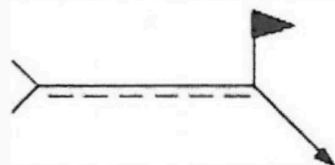
Key

- 1 - elementary symbol (fillet weld)
- 2 - supplementary symbol (concave fillet weld contour, field weld, weld-all-around)
- 3 - complementary information (shielded metal arc welding (SMAW)/process 111 in accordance with ISO 4063)
- 4 - dimensions (5 mm nominal throat thickness intermittent fillet weld, composing 4 weld elements 100 mm in length with 200 mm spacing between elements)
- 5 - tail
- 6a - reference line (continuous)
- 6b - dashed line (system A only)

WELD SYMBOLS		
 SQUARE BUTT WELD 	 SINGLE V BUTT WELD 	 SINGLE V BUTT WELD WITH BROAD ROOT FACE 
 SINGLE BEVEL BUTT WELD 	 SINGLE J BUTT WELD 	 SINGLE U BUTT WELD WITH BROAD ROOT FACE 
 SINGLE V BUTT WELD WITH BACKING RUN 	 FILLET WELD 	 SPOT WELD 

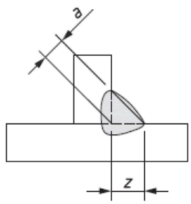
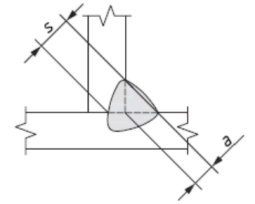


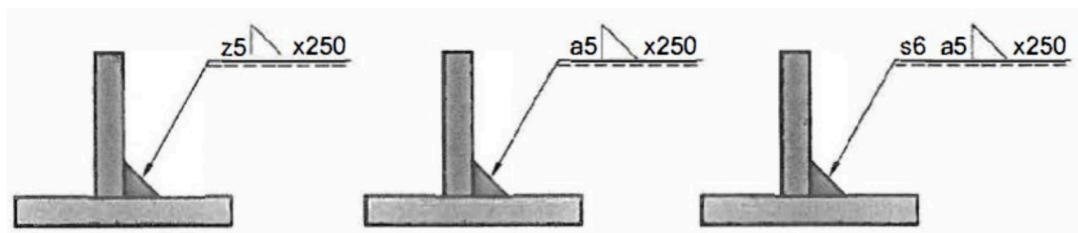
WELD WITH FLAT FACE	WELD WITH CONVEX FACE	WELD WITH CONCAVE FACE
NEAR SIDE 	NEAR SIDE 	NEAR SIDE 
FAR SIDE 	FAR SIDE 	FAR SIDE 

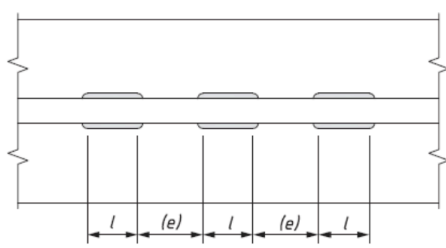
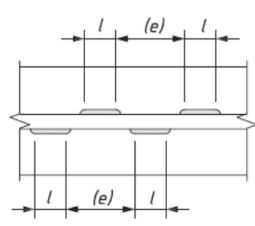
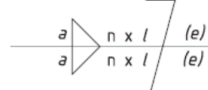
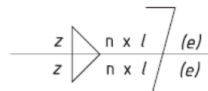
SITE WELD	WELD ALL ROUND	WELD PROCESS
		
		

Dimensioning Welds

DIMENSIONING FILLET WELD

Fillet		 or 
Deep penetration		



Chain intermittent		 or 
Staggered intermittent		 or 

Weld Tolerances

ISO 13920 general tolerances

Welding Processes (ISO 4063)

Arc Welding (1x or 1xx) (most common, heat produced by electric arcs)

11 Metal Arc Welding w/o gas protection

111 Manual Metal arc welding (metal arc welding with covered electrode); shielded metal arc welding

13 gas shielded metal arc welding; gas metal arc welding (GMAW)

131 - MIG (metal inert gas) welding with solid wire electrode; GMAW using inert gas and solid wire electrode

132 - MIG welding with flux cored electrode; Flux cored arc welding

14 gas shielded arc welding with non consumable tungsten electrode; tungsten arc welding

141 - TIG welding with solid filler material,

142 - autogenous TIG welding, autogenous gas tungsten arc welding using inert gas

Resistance Welding (through current producing heat via resistance)

21 - Resistance Spot welding, spot welding

211 - indirect spot welding

212 - direct spot welding

22 - resistance seam welding, seam welding

221 - indirect

222 direct

Gas Welding

31 - Oxy Gas Welding

311 Oxy-acetylene welding

Brazing, Soldering & Braze Welding

brazing/soldering: filler metal is melted at a lower m.p. than parent metal,

soldering uses filler < 450C

brazing uses filler > 450C

parent metal does not melt

braze welding: similar to fusion welding but with filler material, however capillary action of molten metal nor melting of parent metal is done

91 – Brazing with local heating

92 – Brazing with global heating

94 – Soldering with local heating

97 – Weld brazing; Braze welding

Green Engineering

Green Product Life Cycle & Assessment

considers the stages:

Material Extraction

consider resource depletion, water/air pollution, habitat destruction

Material Processing

energy consumption, water/air pollution, waste generation

Manufacturing

energy consumption, greenhouse gases, waste generation, pollution

Distribution and Use

energy, greenhouse gas, waste, pollution

End of Life

waste, resource depletion, pollution

Steps for designing Green products

analyse the product platform and EOL tasks

analyse product platform, disassembly, reuse and recycling tasks,
material/information flow, create action plans and plan potential
improvements

determine degree of disassembly, reusability, recyclability of products &
components

plan return logistics & information flows from production to eol

develop solutions for disassembly, reuse and recycling

prepare guidelines for future green product design

Green Engineering Guidelines

modular product structure
standardisation
material selection
minimise waste and harmful materials
increase ease of separation
product recovery at EOL.